

IS6400 Business Data Analytics

Dr. LIU Junming

City University of Hong Kong

Lecture 4

☐ Quick Review

- ☐ Introduction to Feature Engineering
- ☐ Feature Importance
- ☐ Feature Selection and Dimension Reduction

Reading Material:

☐ HM Chapter 8

Machine Learning Progress

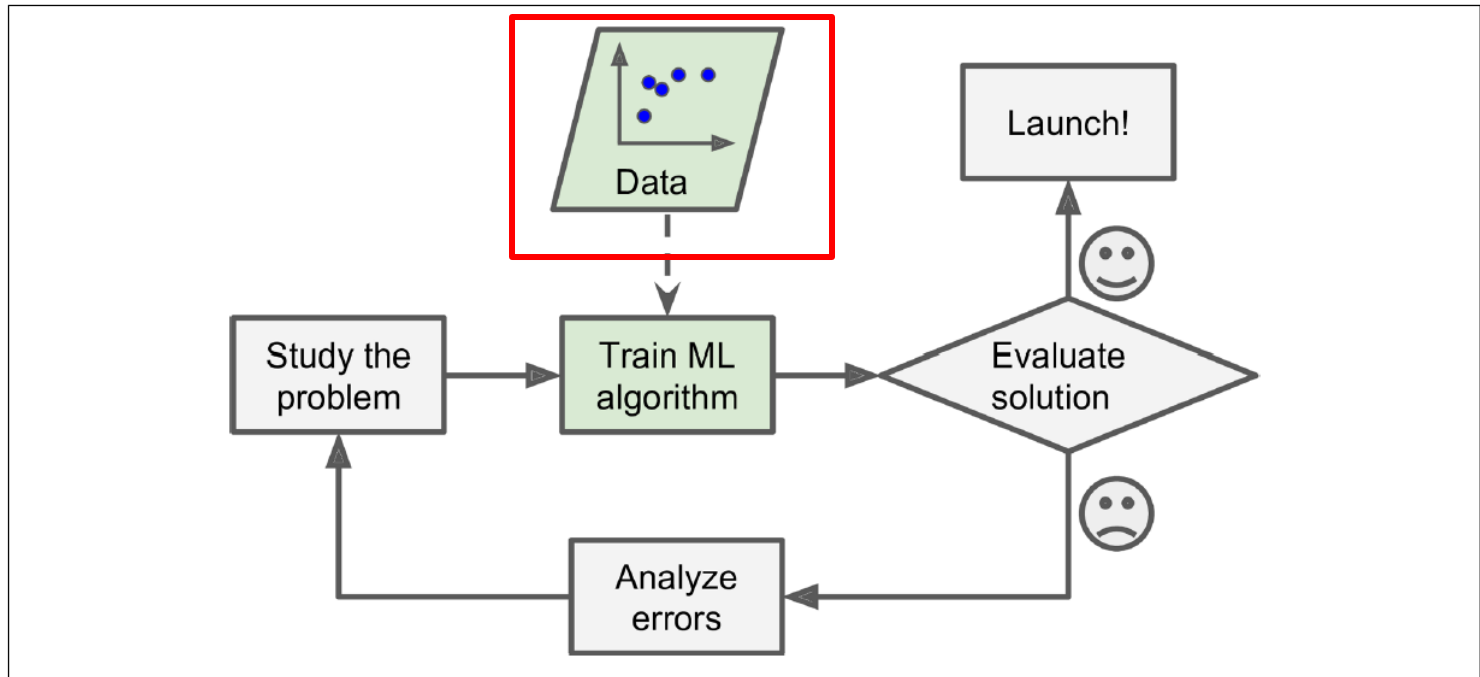


Figure 1-2. Machine Learning approach



1. Transfer the business problem to a machine learning task
2. Study the data related to the problem
3. Select a model and train it using the data
4. Implement the model for future data

What is Data?

- ❖ Collection of data **objects** and their **attributes**
- ❖ An attribute is a property or characteristic of an object
 - ∞ Examples: eye color of a person, temperature, etc.
 - ∞ Attribute is also known as variable, field, characteristic, or feature
- ❖ A collection of attributes describe an object
 - ∞ Object is also known as record, point, case, sample, entity, or instance

Attributes

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

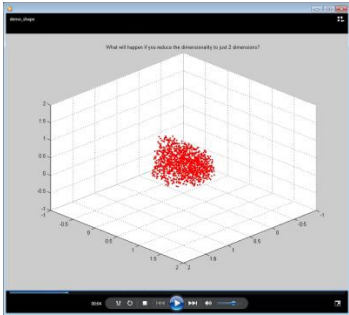
Objects

Important Characteristics of Structured Data

- Data Quality
- Dimensionality
 - ◆ The number of attributes
 - ◆ Curse of Dimensionality
- Sparsity
 - ◆ Only presence counts
 - ◆ Less than 1% of the entries are non-zero.
- Resolution
 - ◆ Patterns depend on the scale
 - ◆ The surface of the Earth seems very uneven at a resolution of a few meters, but is relatively smooth at a resolution of tens of kilometers.

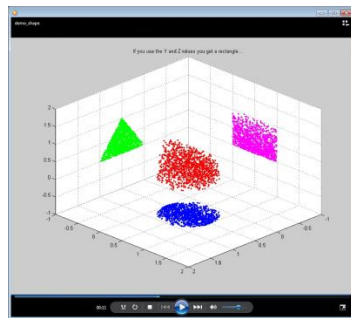
Screen dumps of a short video from www.cs.gmu.edu/~jessica/DimReducDanger.htm
I recommend you imbed the original video instead

A cloud of points in 3D

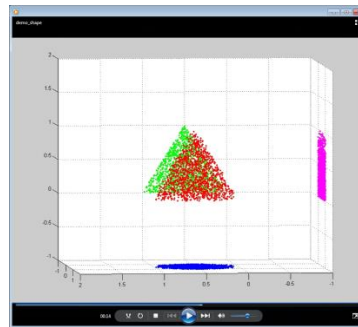


Can be projected into 2D

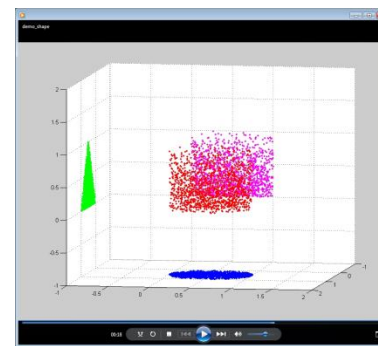
XY or XZ or YZ



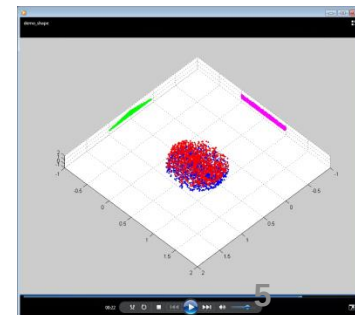
In 2D XZ we see
a triangle



In 2D YZ we see
a square



In 2D XY we see
a circle



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Reading Material:

☒ **HM Chapter 8**

Hands-On Machine Learning with Scikit-Learn and TensorFlow; Concepts, Tools, and Techniques

Introduction

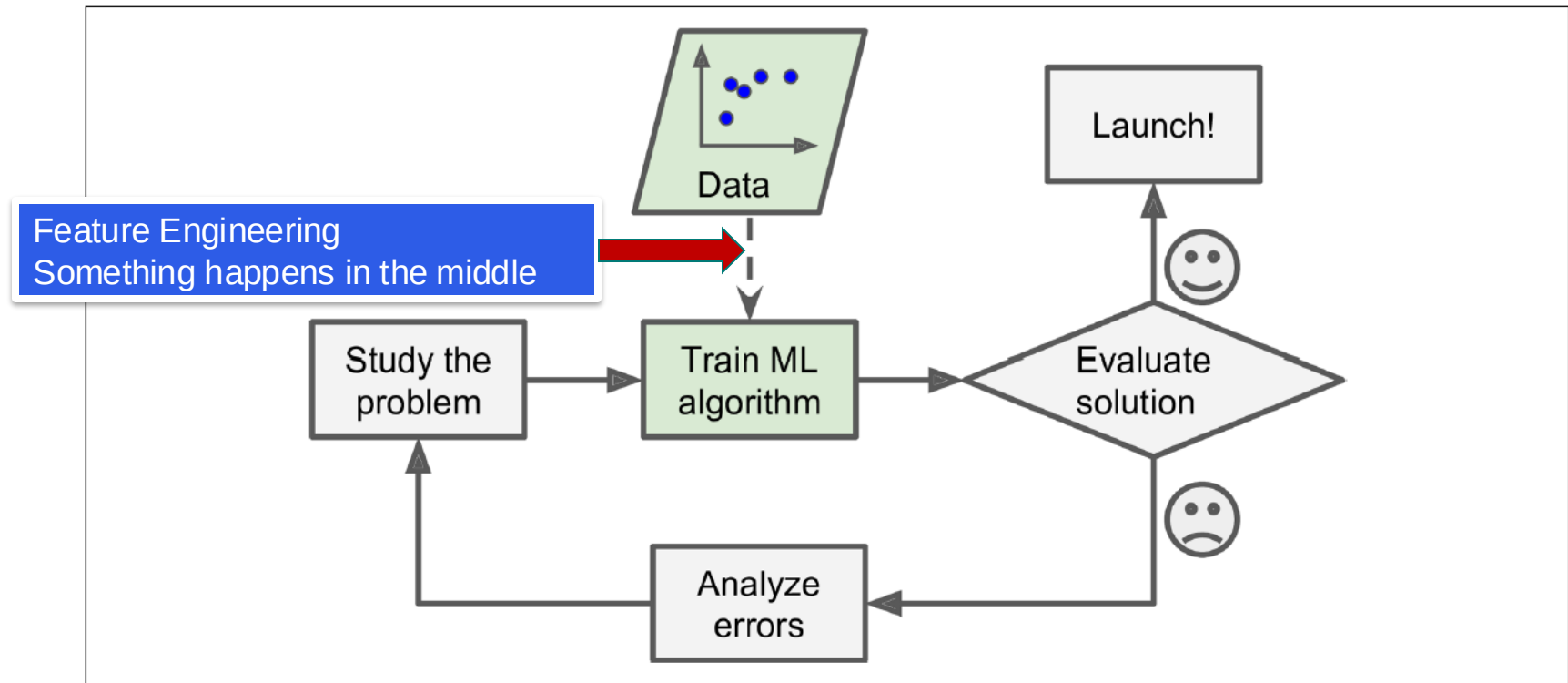


Figure 1-2. Machine Learning approach

Introduction

❖ What is Feature?

☞ A set of attributes to describe a complex object.

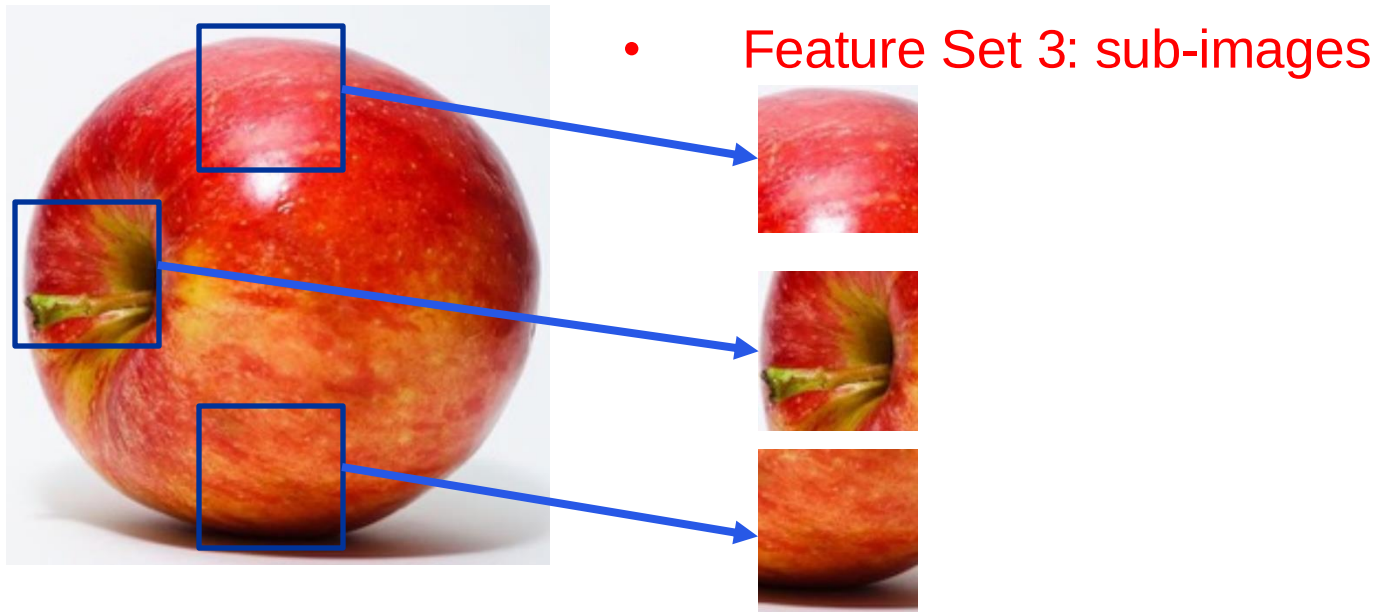


- Feature Set 1
 - Size: 10 cm * 10 cm
- Feature Set 2:
 - Color in RGB: 220,156,140

Introduction

❖ What is Feature?

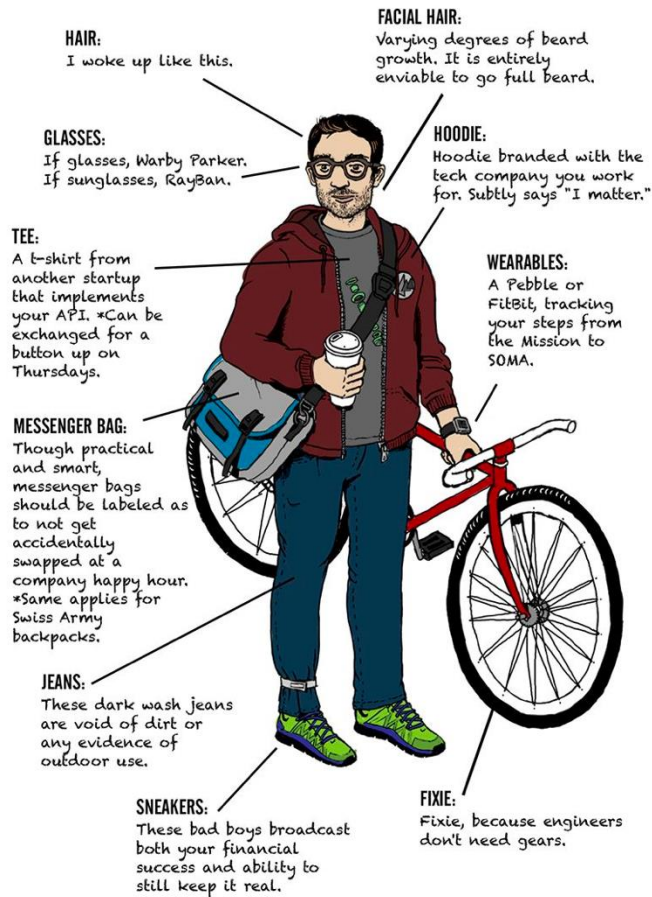
☞ A set of attributes to describe a complex object.



Introduction

readwrite presents

THE TECH UNIFORM



Feature of a programmer?

Introduction

❖ How to describe a document?

EX: For example, your **spam** filter is a **Machine Learning** program that can learn to flag **spam** given examples of **spam emails** (e.g., flagged by users) and examples of regular (non**spam**, also called “ham”) **emails**. The examples that the system uses to learn are called the training set. Each training example is called a training instance (or sample). In this case, the task T is to flag **spam** for new **emails**, the experience E is the training data, and the performance measure P needs to be defined; for example, you can use the ratio of correctly **classified emails**. This particular performance measure is called accuracy and it is often used in **classification** tasks.

Key	spam	machine	email	class	tree
Doc1	5	1	4	2	0

Introduction

❖ How to profile a social media user?



User	Followers	Role	Gender	Tag
Angel				
Austin				

Introduction

- ❖ Feature Engineering: select/create the most representative features.



Feature set 1

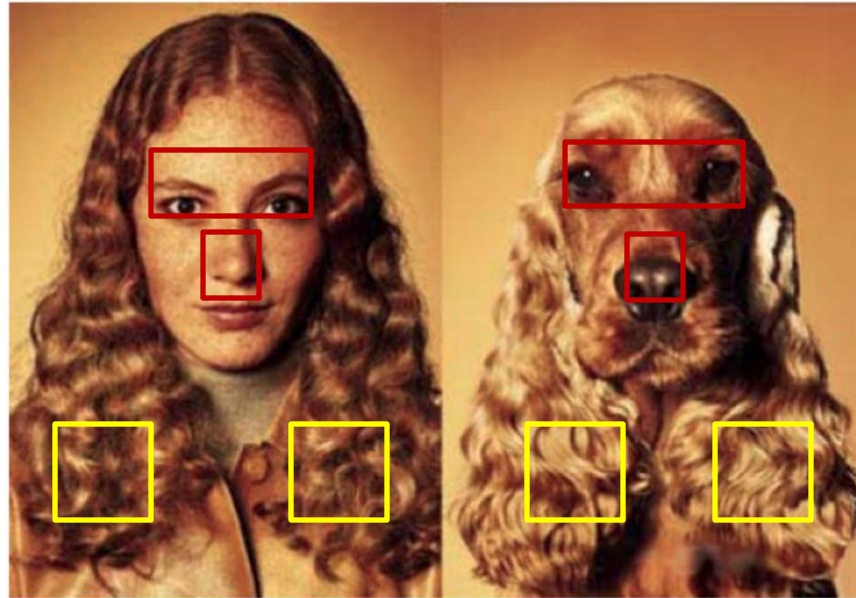
Object	Feature1: length	Feature 2: width
Apple	10 cm	10 cm
Orange	10 cm	10 cm

Feature set 2

Object	RGB 1	RGB 2	RGB 3
Apple	220	156	140
Orange	243	194	113

Which Feature set is better?
The most representative

Introduction



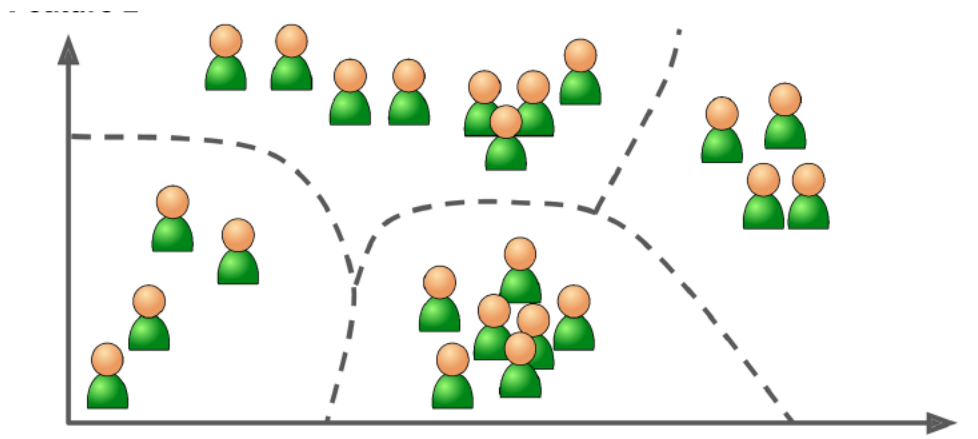
Which feature set is better?

Introduction

❖ Feature Engineering is Problem-Specific

☞ Task 1: find potential sport-game customers

☞ Task 2: find potential Covid-19 related news subscribers



Feature Engineering Challenges

- ❖ Create/Choose Features from Raw Data
- ❖ Duplicate/Redundant Feature
- ❖ Curse of dimensionality

Create Representative Features

- ❖ Create Representative from Raw Data
- ❖ What are the representative keywords?

EX: For example, your **spam** filter is a **Machine Learning** program that can learn to flag **spam** given examples of **spam emails** (e.g., flagged by users) and examples of regular (non **spam**, also called “ham”) **emails**. The examples that the system uses to learn are called the training set. Each training example is called a training instance (or sample). In this case, the task T is to flag **spam** for new **emails**, the experience E is the training data, and the performance measure P needs to be defined; for example, you can use the ratio of correctly **classified emails**. This particular performance measure is called accuracy and it is often used in **classification** tasks.

Duplicate/Redundant Feature

❖ Two Features:

❧ Temperature

❧ Feeling: Cold – Warm - Hot

Is feature “Feeling” useful if we already have feature “temperature”?

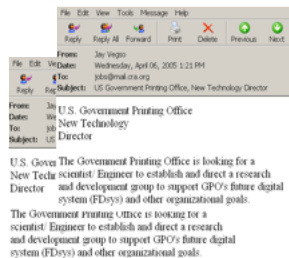
“Feeling” is redundant.

Curse of Dimension

❖ Is High dimension always good?

Web Pages

Emails



Documents

D_1
 D_2
 $\vdots$
 D_M

Terms

T_1 T_2 T_N

C

12	0		6
3	10		28
$\vdots$	$\vdots$		$\vdots$
0	11		16

Sports
Travel
 $\vdots$
Jobs

Internet

ACM Portal

IEEE Xplore

PubMed

Digital Libraries

- **Task:** To classify unlabeled documents into categories
- **Challenge:** thousands of terms
- **Solution:** to apply dimensionality reduction

Curse of Dimension

❖ Is High dimension always good?

❧ Large Data Space and Storage Requirement

❧ Cause Sparsity

Practice

If we have 1,000,000 data with 100,000 dimensions, how much memory do we need?

$$\begin{aligned}\text{Ans:} & 10^6 \times 10^5 \times 8 \\ &= 8 \times 10^{11} \text{ (B)} \\ &= 800 \text{ (GB)}\end{aligned}$$

Noisy Feature

Is every feature useful?
Redundancy?

Theory

Without any assumption, you need $O(\frac{1}{\epsilon^d})$ data to achieve ϵ error for d-dimensional data

Feature Engineering

- ❖ Select the most representative feature set from raw data.

Trade-off between information, data space, and computation time

- CIFAR-10 image classification with **raw pixels** as features and using approximated kernel SVM



(Li and Póczos, Utilize Old Coordinates: Faster Doubly Stochastic Gradients for Kernel Methods, UAI 2016)

Dimensions	Accuracy	Time
3072 (all)	63.1%	~2 Hrs
100 (PCA)	59.8%	250 s

Trade-off between information, space and **time**

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Reading Material:

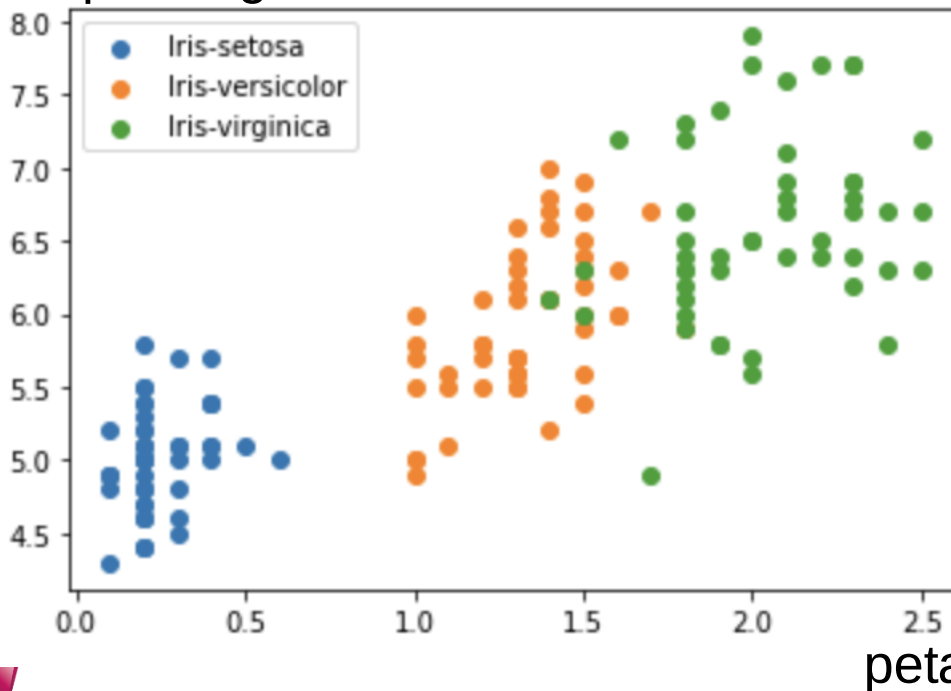
- ☒ **HM Chapter 8**

Feature Importance

❖ Feature importance:

Techniques that calculate a score for all the input features to measure the representativeness of different features

Sepal length



How to measure the importance of “petal width” and “sepal length” in this classification task?

THEN, the machine can use this “importance score” to automatically select the most important feature.

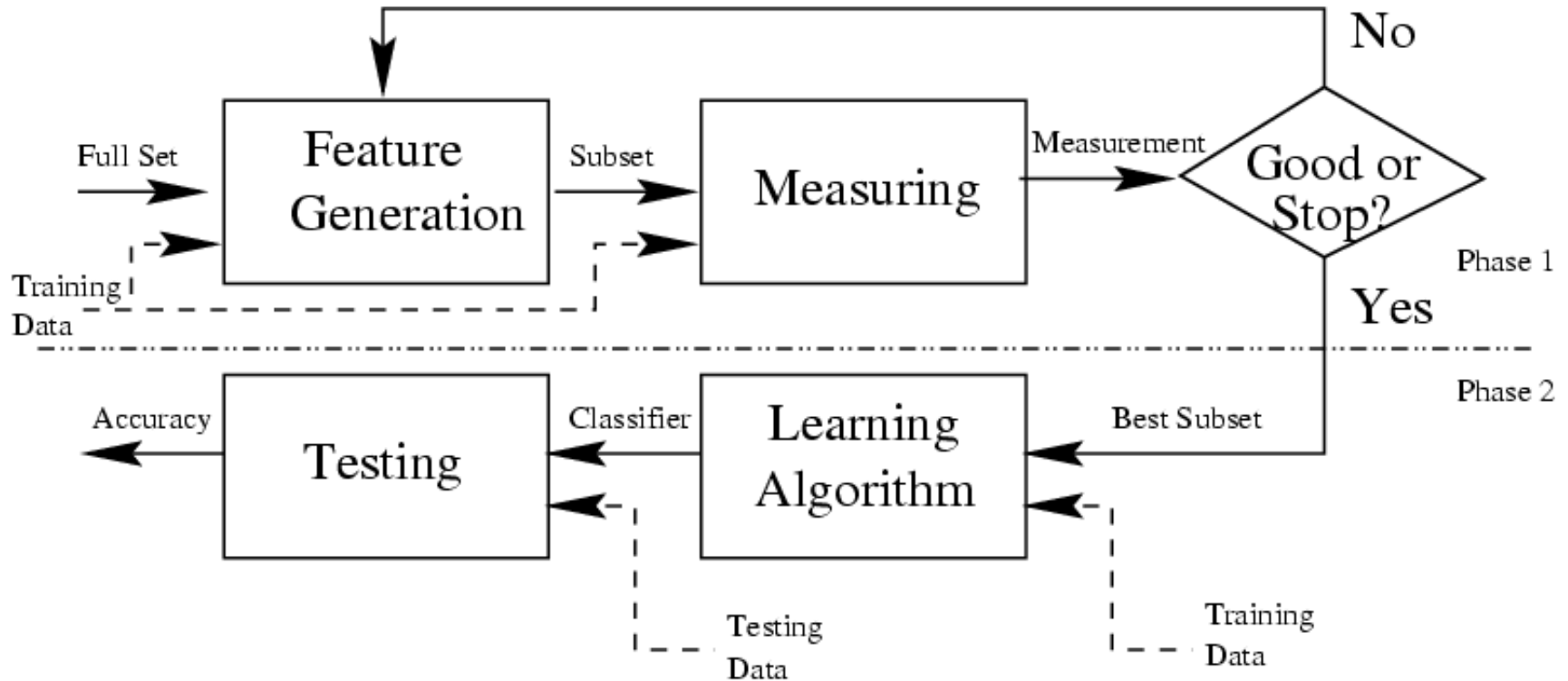
Evaluation Measures for Ranking and Selecting Features

- ❖ The goodness of a feature/feature subset is dependent on measures
- ❖ Various measures
 - ∞ Information measures (Yu & Liu 2004, Jebara & Jaakkola 2000)
 - ∞ Distance measures (Robnik & Kononenko 03, Pudil & Novovicov 98)
 - ∞ Dependence measures (Hall 2000, Modrzejewski 1993)
 - ∞ Consistency measures (Almuallim & Dietterich 94, Dash & Liu 03)
 - ∞ Accuracy measures (Dash & Liu 2000, Kohavi&John 1997)

Evaluation Measures for Ranking and Selecting Features

- ❖ The goodness of a feature/feature subset is dependent on measures
- ❖ Various measures
 - ∞ Filter: Without learning algorithm involvement
 - ∞ Wrapper: With learning algorithm involvement
- ❖ Same objective: select the optimal feature set.

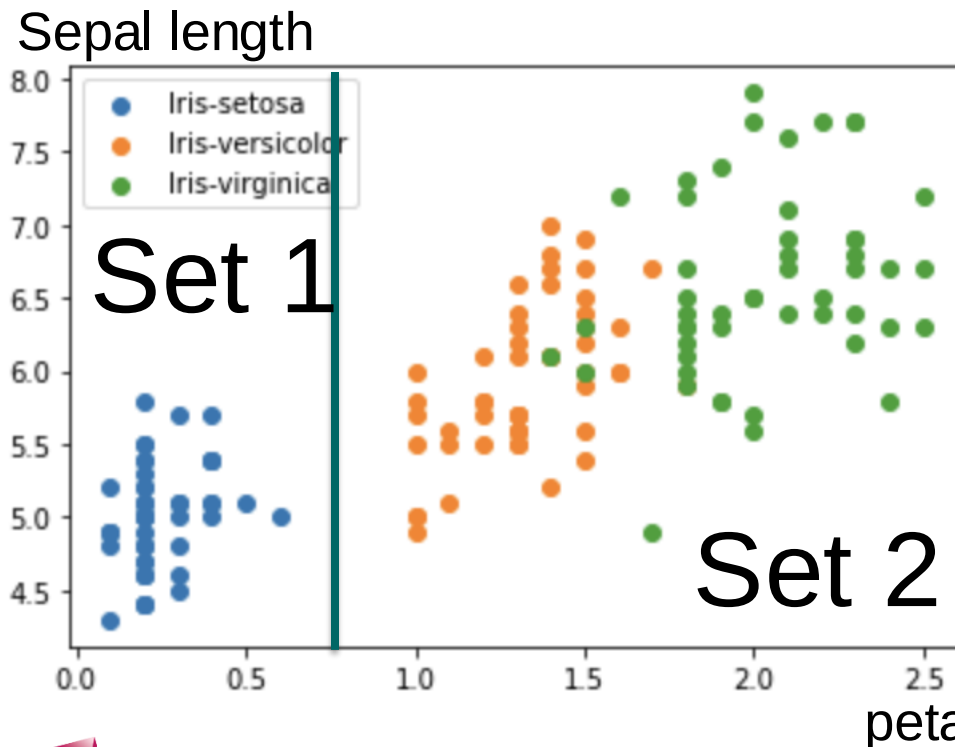
Filter Model



Filter model: class purity

- ❖ Filter: Without learning algorithm involvement

A simple measurement: class purity



1. User petal width
Separate the data into
two sets:

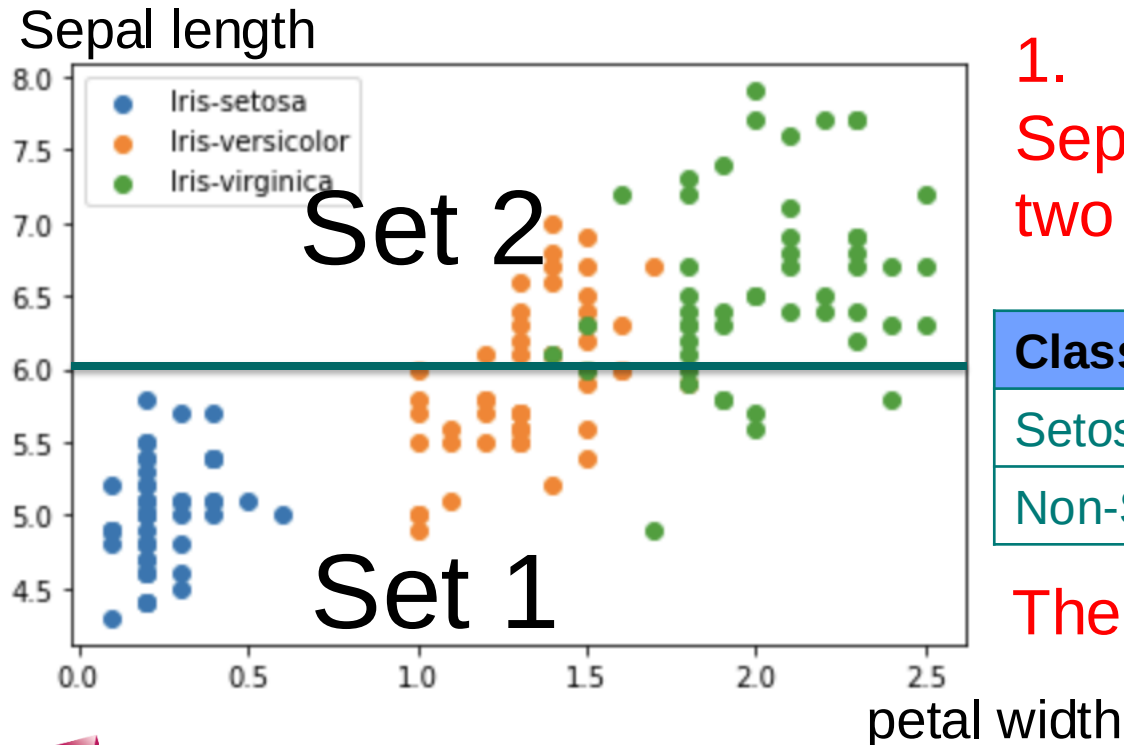
Class	Set 1	Set 2
Setosa	50	0
Non-Setosa	0	100

The two sets are pure!

Filter model: class purity

- ❖ Filter: Without learning algorithm involvement

A simple measurement: class purity



1. User sepal length
Separate the data into
two sets:

Class	Set 1	Set 2
Setosa	50	0
Non-Setosa	10	90

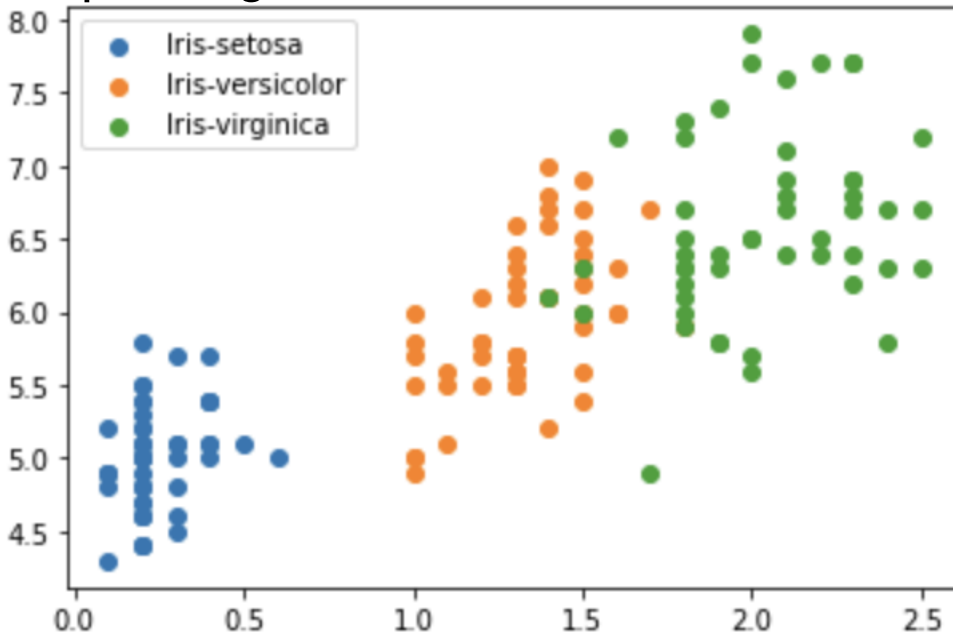
The two sets are not pure!

Filter model: class purity

❖ Filter: Without learning algorithm involvement

A simple measurement: class purity

Sepal length



Class purity (simple counting of distinctiveness)

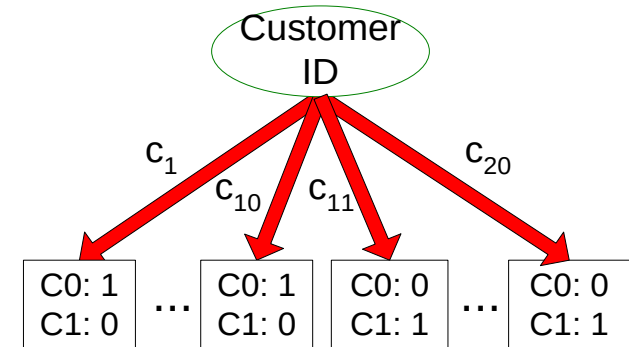
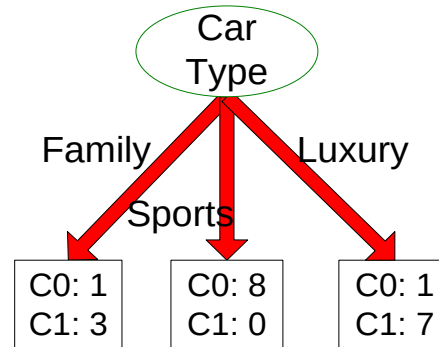
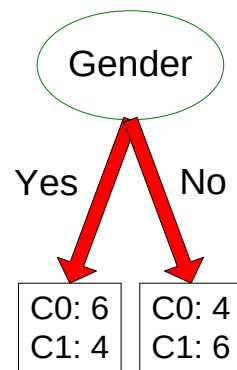
Class	Set 1	Set 2
Setosa	n10	n20
Non-Setosa	n11	n21

The purer, the better.

Feature importance measurement

Before Splitting: 10 records of class 0,
10 records of class 1

Customer Id	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1



Which test condition is the best?

How to determine the importance

❖ Greedy approach:

❧ Features with **purser** class distribution are preferred

❖ Need a measure of node impurity:

C0: 5
C1: 5

High degree of impurity

C0: 9
C1: 1

Low degree of impurity

Measures of Node Impurity

❖ Gini Index

$$GINI(t) = 1 - \sum_j [p(j | t)]^2$$

❖ Entropy

$$Entropy(t) = -\sum_j p(j | t) \log p(j | t)$$

❖ Univariate Score: $-Log(p_{value})$

Feature importance

1. Compute impurity measure (P) before splitting
2. Compute impurity measure (M) after splitting
 - Compute impurity measure of each child node
 - Compute the average impurity of the children (M)
3. Choose the attribute test condition that produces the highest gain

$$\text{Gain} = P - M$$

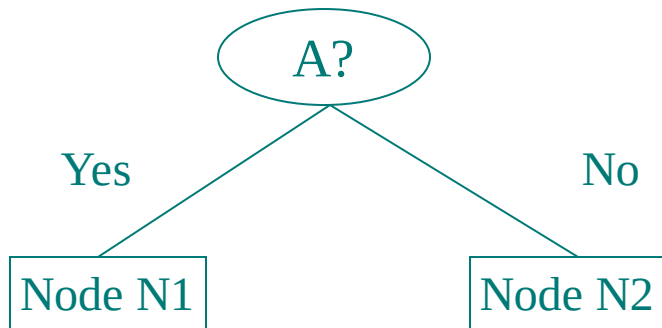
or equivalently, lowest impurity measure after splitting (M)

Feature importance

Before Splitting:

C0	N00
C1	N01

→ P



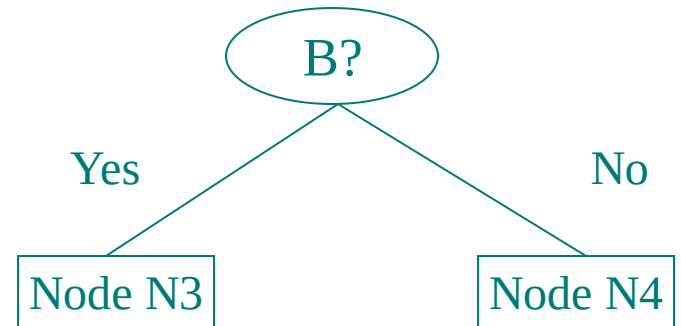
C0	N10
C1	N11

C0	N20
C1	N21

↓
M11

↓
M12

M1



C0	N30
C1	N31

C0	N40
C1	N41

↓
M21

↓
M22

M2

$$\text{Gain} = P - M1 \quad \text{vs} \quad P - M2$$

Measure of Impurity: GINI

❖ Gini Index for a given node t :

$$GINI(t) = 1 - \sum_j [p(j | t)]^2$$

(NOTE: $p(j | t)$ is the relative frequency of class j at node t).

- ❧ Maximum ($1 - 1/n_c$) when records are equally distributed among all classes, implying least interesting information
- ❧ Minimum (0.0) when all records belong to one class, implying most interesting information

C1	0
C2	6
Gini=0.000	

C1	1
C2	5
Gini=0.278	

C1	2
C2	4
Gini=0.444	

C1	3
C2	3
Gini=0.500	

Computing Gini Index of a Single Node

$$GINI(t) = 1 - \sum_j [p(j | t)]^2$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$Gini = 1 - P(C1)^2 - P(C2)^2 = 1 - 0 - 1 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$Gini = 1 - (1/6)^2 - (5/6)^2 = 0.278$$

C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$Gini = 1 - (2/6)^2 - (4/6)^2 = 0.444$$

Computing Gini Index for a Collection of Nodes

- ❖ When a node p is split into k partitions (children)

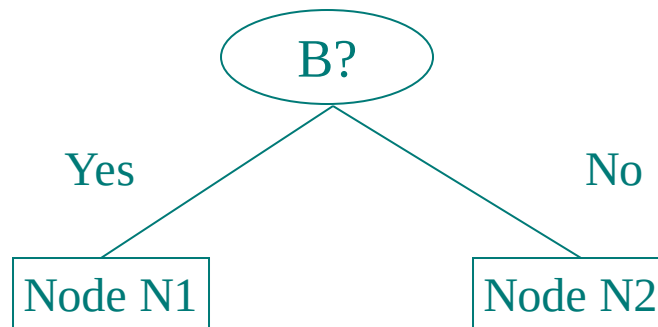
$$GINI_{split} = \sum_{i=1}^k \frac{n_i}{n} GINI(i)$$

where, n_i = number of records at child i ,
 n = number of records at parent node p .

- ❖ Choose the attribute that minimizes weighted average Gini index of the children
- ❖ Gini index is used in decision tree algorithms such as CART, SLIQ, SPRINT

Binary Attributes: Computing GINI Index

- Splits into two partitions
- Effect of Weighing partitions:
 - Larger and Purer Partitions are sought for.



	Parent
C1	7
C2	5
Gini = 0.486	

$$\begin{aligned} \text{Gini}(N1) &= 1 - (5/6)^2 - (1/6)^2 \\ &= 0.278 \end{aligned}$$

$$\begin{aligned} \text{Gini}(N2) &= 1 - (2/6)^2 - (4/6)^2 \\ &= 0.444 \end{aligned}$$

	N1	N2
C1	5	2
C2	1	4
Gini=0.361		

$$\begin{aligned} \text{Gini(Children)} &= 6/12 * 0.278 + \\ &\quad 6/12 * 0.444 \\ &= 0.361 \end{aligned}$$

Categorical Attributes: Computing Gini Index

- ❖ For each distinct value, gather counts for each class in the dataset
- ❖ Use the count matrix to make decisions

Multi-way split

	CarType		
	Family	Sports	Luxury
C1	1	8	1
C2	3	0	7
Gini	0.163		

Two-way split
(find best partition of values)

	CarType	
	{Sports, Luxury}	{Family}
C1	9	1
C2	7	3
Gini	0.468	

	CarType	
	{Sports}	{Family, Luxury}
C1	8	2
C2	0	10
Gini	0.167	

Measure of Impurity: Entropy

❖ Entropy at a given node t :

$$Entropy(t) = -\sum_j p(j | t) \log p(j | t)$$

(NOTE: $p(j | t)$ is the relative frequency of class j at node t).

- ❖ Maximum ($\log n_c$) when records are equally distributed among all classes implying least information
- ❖ Minimum (0.0) when all records belong to one class, implying most information

∞ Entropy based computations are quite similar to the GINI index computations

Computing Entropy of a Single Node

$$Entropy(t) = -\sum_j p(j | t) \log_2 p(j | t)$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$Entropy = -0 \log 0 - 1 \log 1 = -0 - 0 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$Entropy = - (1/6) \log_2 (1/6) - (5/6) \log_2 (5/6) = 0.65$$

C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$Entropy = - (2/6) \log_2 (2/6) - (4/6) \log_2 (4/6) = 0.92$$

Computing Information Gain After Splitting

❖ Information Gain:

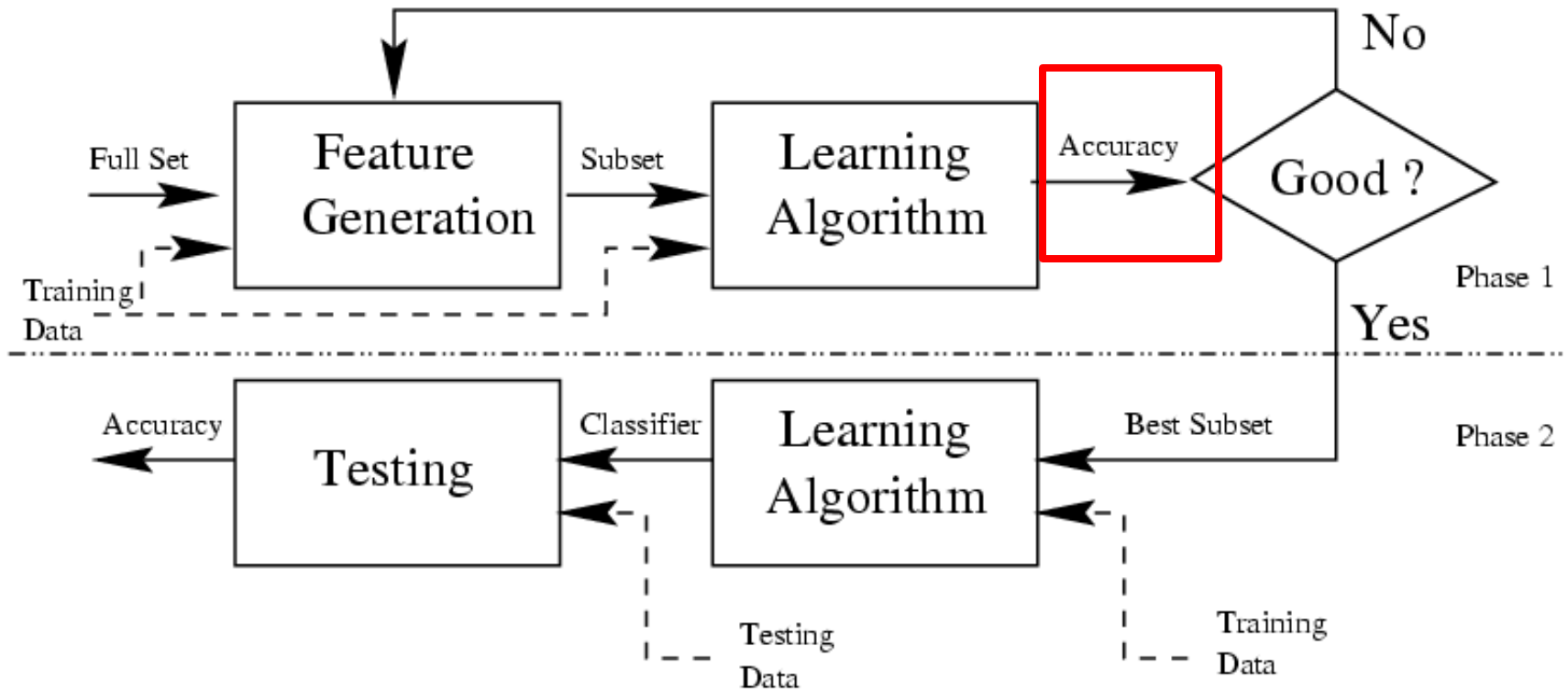
$$GAIN_{split} = Entropy(p) - \left(\sum_{i=1}^k \frac{n_i}{n} Entropy(i) \right)$$

Parent Node, p is split into k partitions;

n_i is number of records in partition i

- ☞ Choose the split that achieves most reduction (maximizes GAIN)
- ☞ Used in the ID3 and C4.5 decision tree algorithms

Wrapper Model



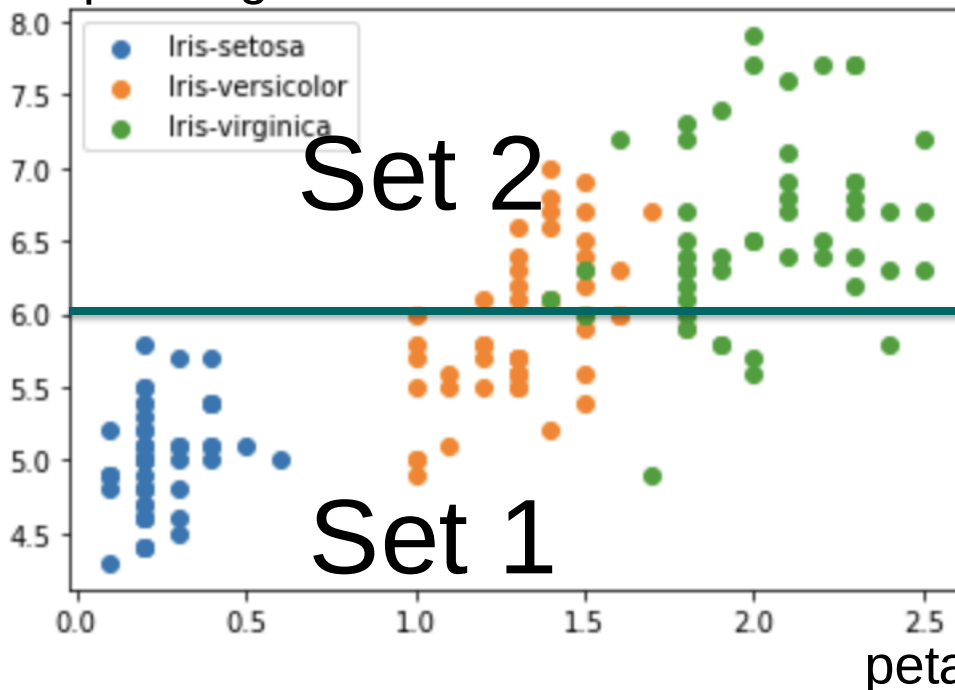
Step 1: use a pre-determined learning model to input the feature

Step 2: use the model performance as feature importance score

In-class Quiz

❖ Calculate the Gini Index after split

Sepal length



1. User sepal length
Separate the data into
two sets:

Class	Set 1	Set 2
Setosa	50	0
Non-Setosa	10	90

$$GINI(t) = 1 - \sum_j [p(j|t)]^2$$

$$GINI_{split} = \sum_{i=1}^k \frac{n_i}{n} GINI(i)$$

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Major Techniques of Dimensionality Reduction

❖ Feature selection

- ❧ Definition

- ❧ Objectives

❖ Feature reduction

- ❧ Definition

- ❧ Objectives

❖ Differences between the two techniques

Feature Selection

❖ Definition

- ❧ A process that chooses an optimal subset of features according to a objective function

❖ Objectives

- ❧ To reduce dimensionality and remove noise
- ❧ To improve mining performance
 - ❖ Speed of learning
 - ❖ Predictive accuracy
 - ❖ Simplicity and comprehensibility of mined results

Feature Reduction

- ❖ Feature reduction refers to the **mapping of the original high-dimensional data onto a lower-dimensional space** $\{x_1, x_2, \dots, x_n\}$

- ❖ Given a set of data points of p variables

Compute their low-dimensional representation:

$$x_i \in \mathbb{R}^d \rightarrow y_i \in \mathbb{R}^p \ (p \ll d)$$

- ❖ Criterion for feature reduction can be different based on different problem settings.

- ∞ Unsupervised setting: minimize the information loss

- ∞ Supervised setting: maximize the class discrimination

Feature Reduction vs. Feature Selection

❖ Feature reduction

- ❧ All original features are used
- ❧ The transformed features are linear combinations of the original features

❖ Feature selection

- ❧ Only a subset of the original features are selected

❖ Continuous versus discrete

Models of Feature Selection

❖ Filter model

- ❧ Separating feature selection from model learning
- ❧ Relying on general characteristics of data (*information, distance, dependence, consistency*)
- ❧ No bias toward any learning algorithm, fast

❖ Wrapper model

- ❧ Relying on a predetermined machine learning model
- ❧ Using predictive accuracy as goodness measure
- ❧ High accuracy, computationally expensive

Feature Selection Basics

- ❖ Definitions of subset optimality
- ❖ Perspectives of feature selection
 - ∞ Subset search and feature ranking
 - ∞ Feature/subset evaluation measures
 - ∞ Models: filter vs. wrapper
 - ∞ Results validation and evaluation

An Example for Optimal Subset

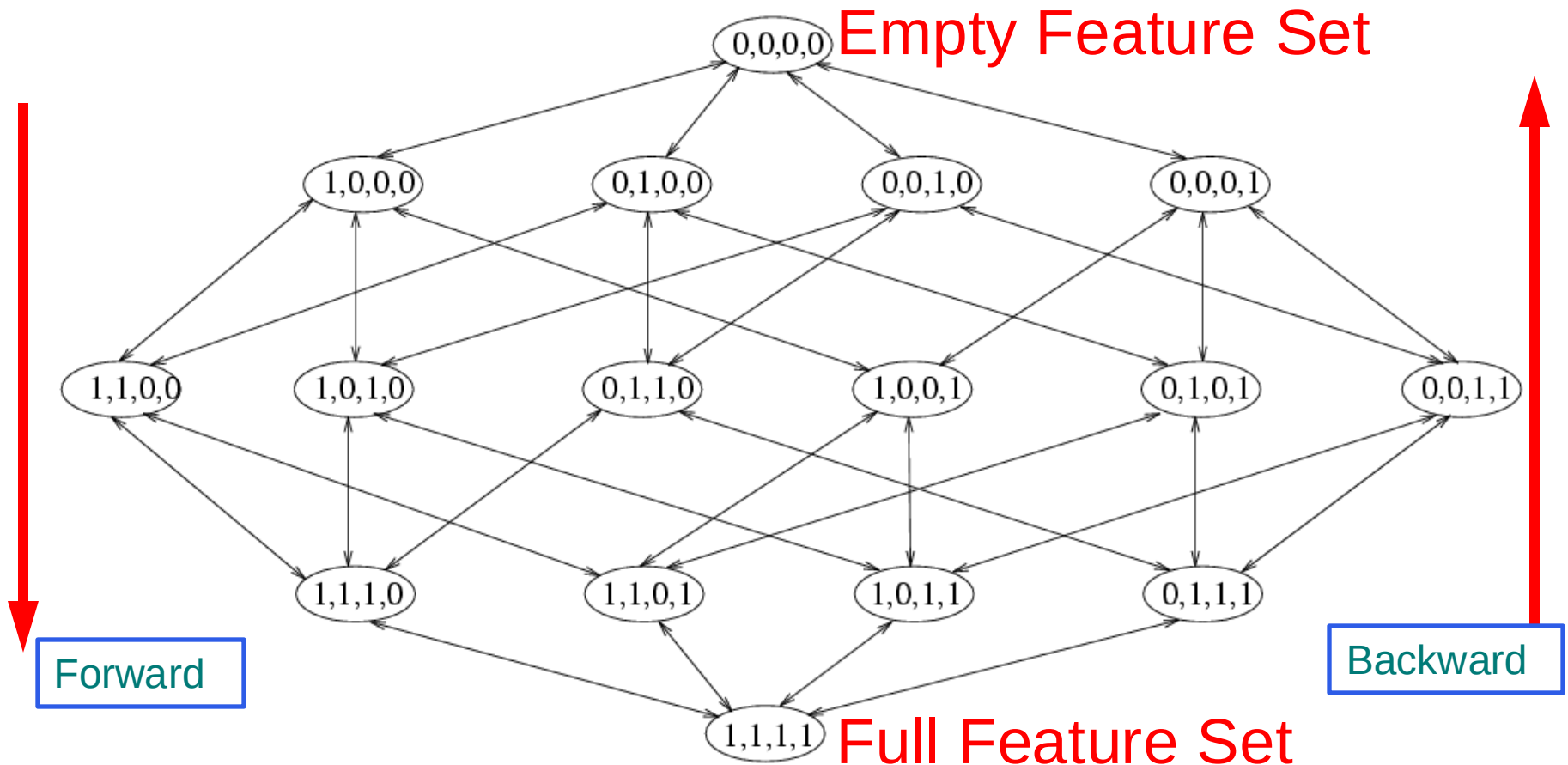


F ₁	F ₂	F ₃	F ₄	F ₅	C
0	0	1	0	1	0
0	1	0	0	1	1
1	0	1	0	1	1
1	1	0	0	1	1
0	0	1	1	0	0
0	1	0	1	0	1
1	0	1	1	0	1
1	1	0	1	0	1

- ❖ Data set (whole set)
 - ∞ Five Boolean features
 - ∞ $C = F_1 \vee F_2$
 - ∞ $F_3 = \neg F_2$, $F_5 = \neg F_4$
 - ∞ Optimal subset:
 $\{F_1, F_2\}$ or $\{F_1, F_3\}$
- ❖ Combinatorial nature of searching for an optimal subset

A Subset Search Problem

❖ An example of search space (Kohavi & John 1997)

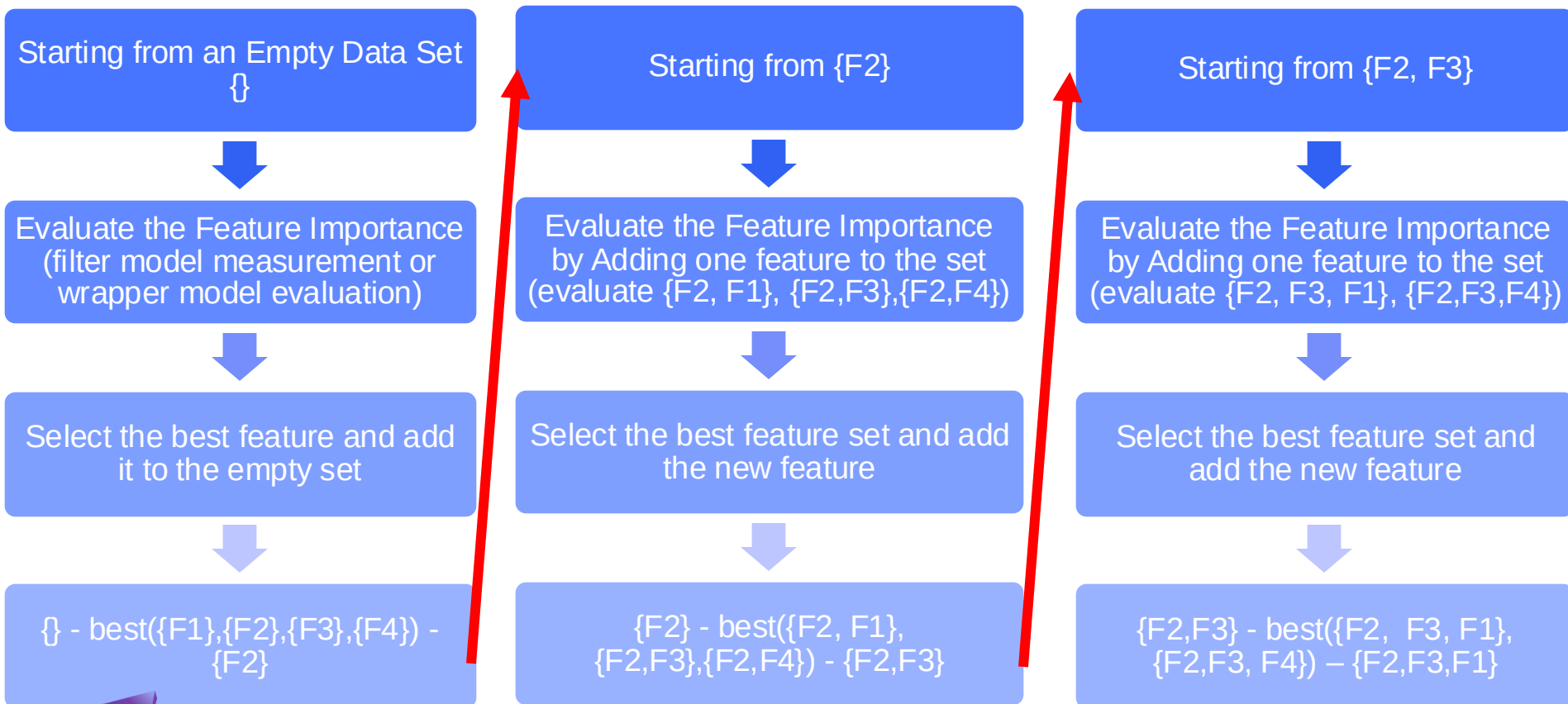


How to make feature selection efficient??

Recursive Feature Selection

- ❖ Can be implemented on Filter Model and Wrapper Model

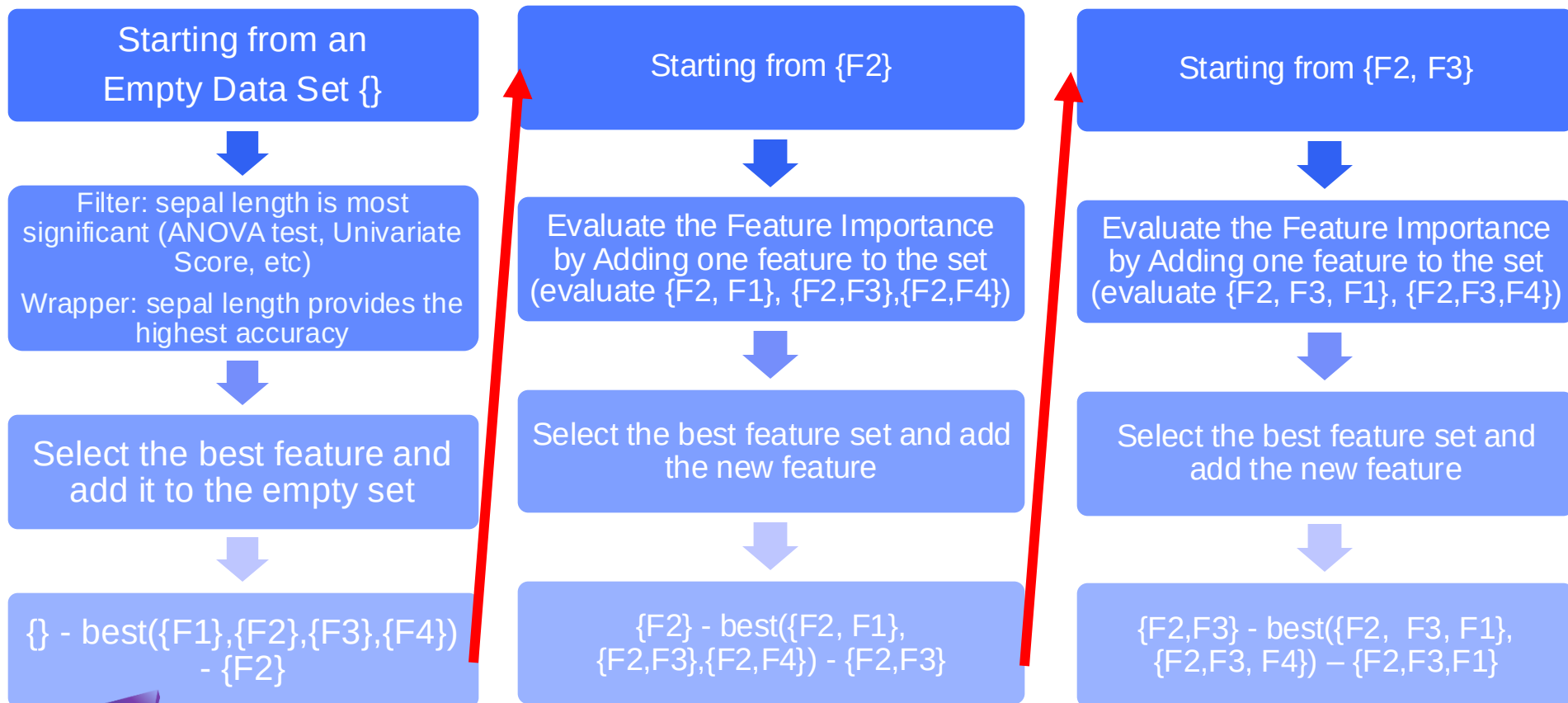
Full set: $\{F1, F2, F3, F4\}$



Evaluate $\{F2\}, \{F2, F3\}, \{F2, F3, F1\}, \{F2, F3, F1, F4\}$

Recursive Feature Selection

- ❖ Can be implemented on Filter Model and Wrapper Model
- Iris data: {Sepal width, sepal length, petal width, petal length}



Feature Reduction: Principal Component Analysis

- ❖ Principal component analysis (PCA)
 - ∞ Reduce the dimensionality of a data set by finding a new set of variables, smaller than the original set of variables
 - ∞ Retains most of the sample's information.
- ❖ By information we mean the variation present in the sample, given by the correlations between the original variables.
 - ∞ The new variables, called principal components (PCs), are **uncorrelated**, and are ordered by the fraction of the total information each retains.

PCA

Find function/operation G to

Map $x \in \mathbb{R}^d$ to $z \in \mathbb{R}^p, p \ll d$

$$z = Gx$$

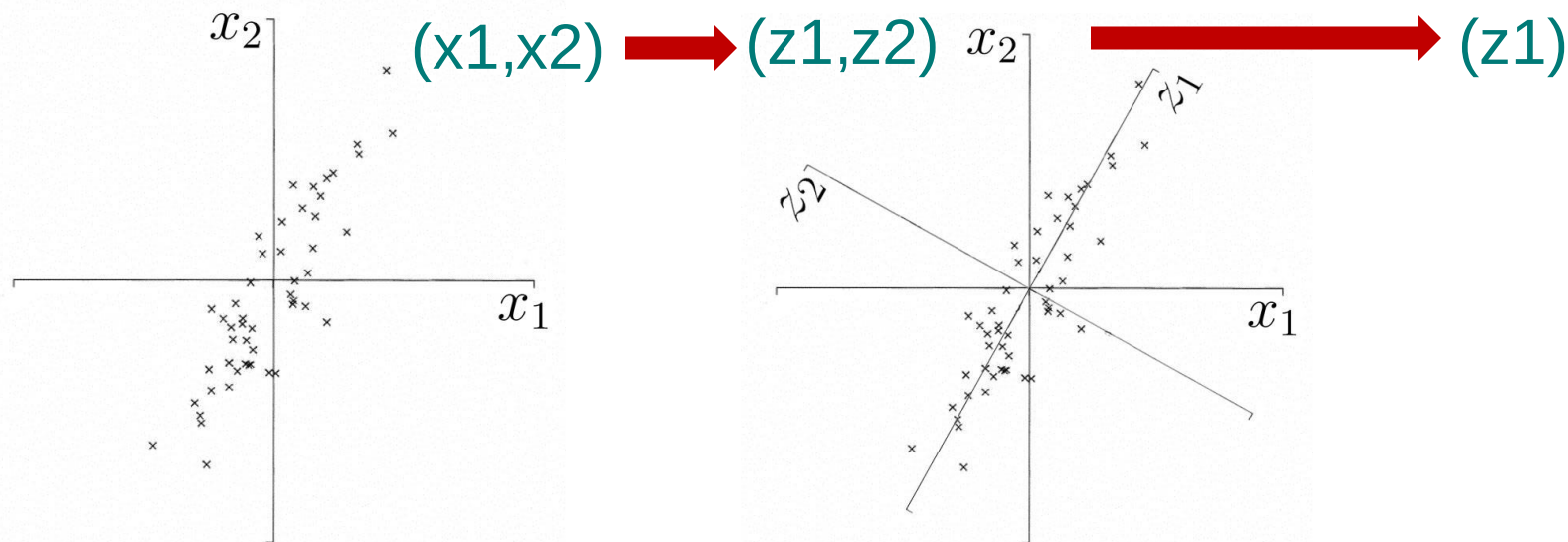
The matrix G consisting of the first p eigenvectors of the covariance matrix S solves the following min problem:

$$\min_{G \in \mathbb{R}^{d \times p}} \|X - G(G^T X)\|_F^2 \text{ subject to } G^T G = I_p$$


$$\|X - \bar{X}\|_F^2$$

reconstruction error

Geometric Picture of Principal Components (PCs)

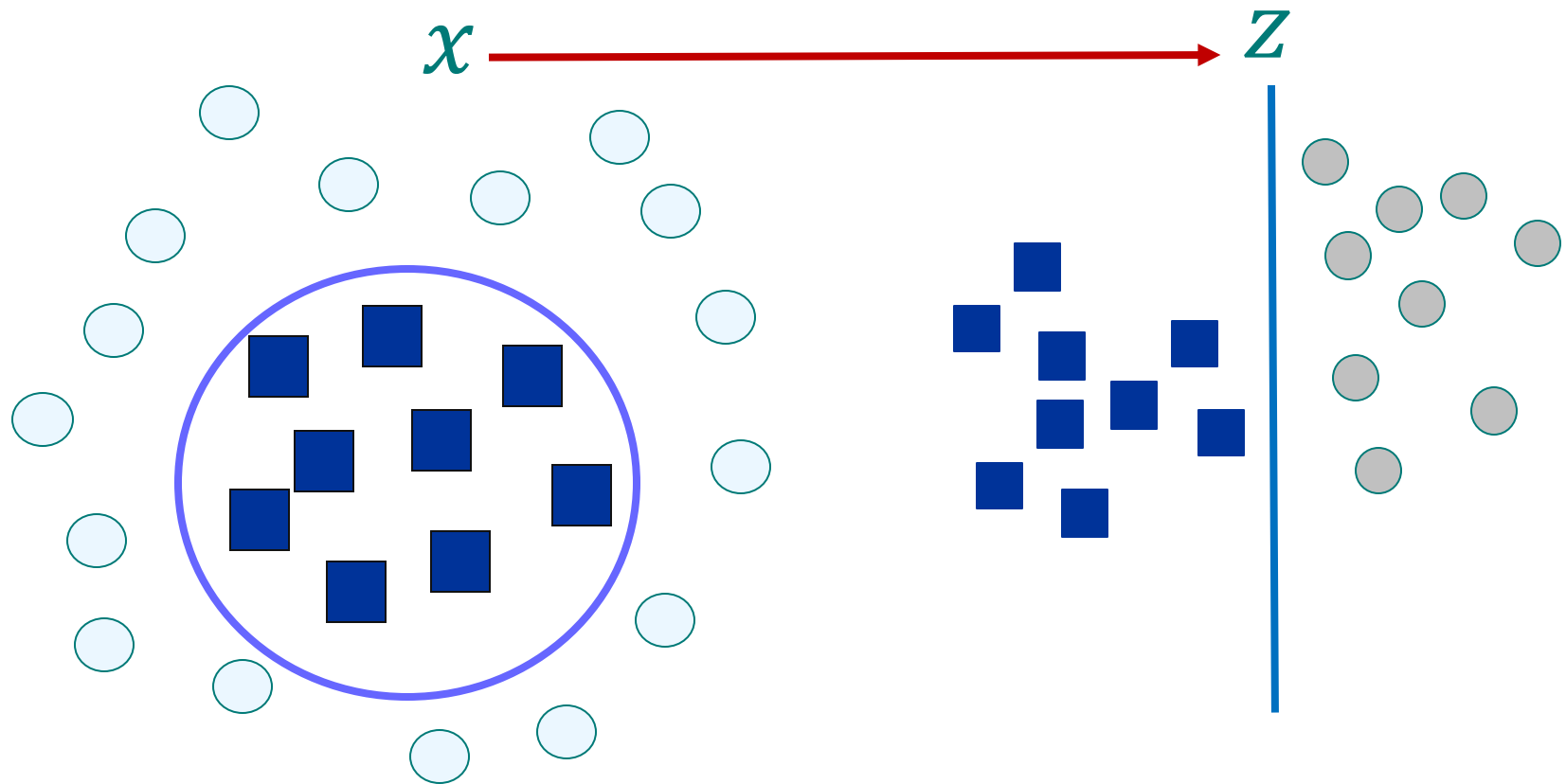


- the 1st PC z_1 is a minimum distance fit to a line in X space
- the 2nd PC z_2 is a minimum distance fit to a line in the plane perpendicular to the 1st PC

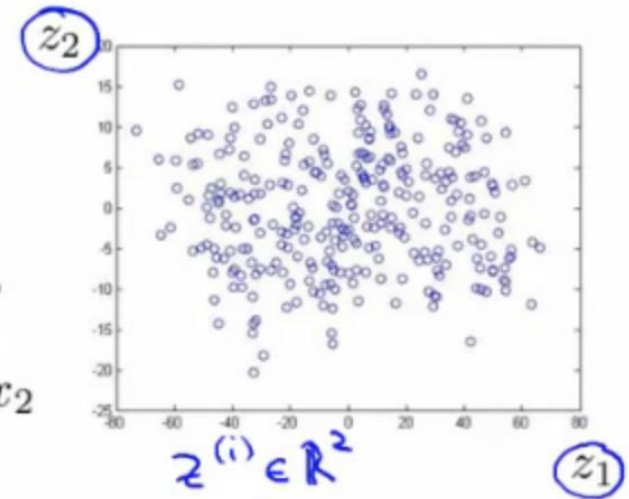
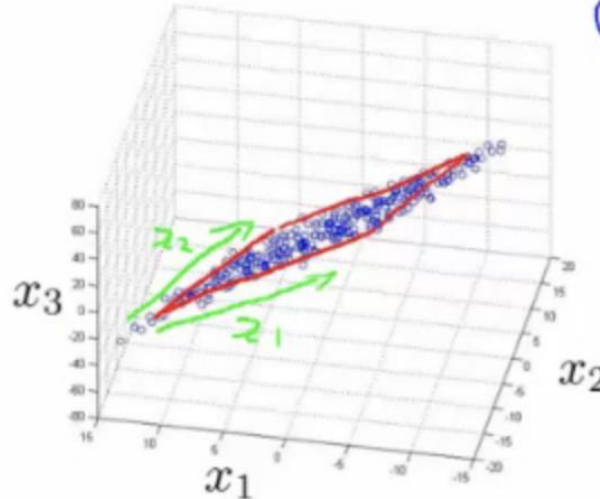
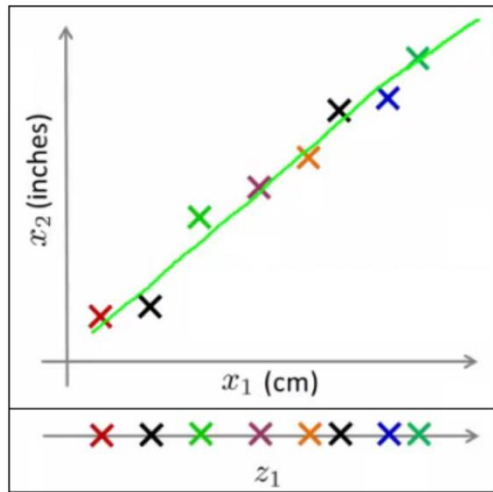
PCs are a series of linear least squares fits to a sample, each orthogonal to all the previous.

Motivation for Non-linear PCA using Kernels

Linear projections will not detect the pattern.



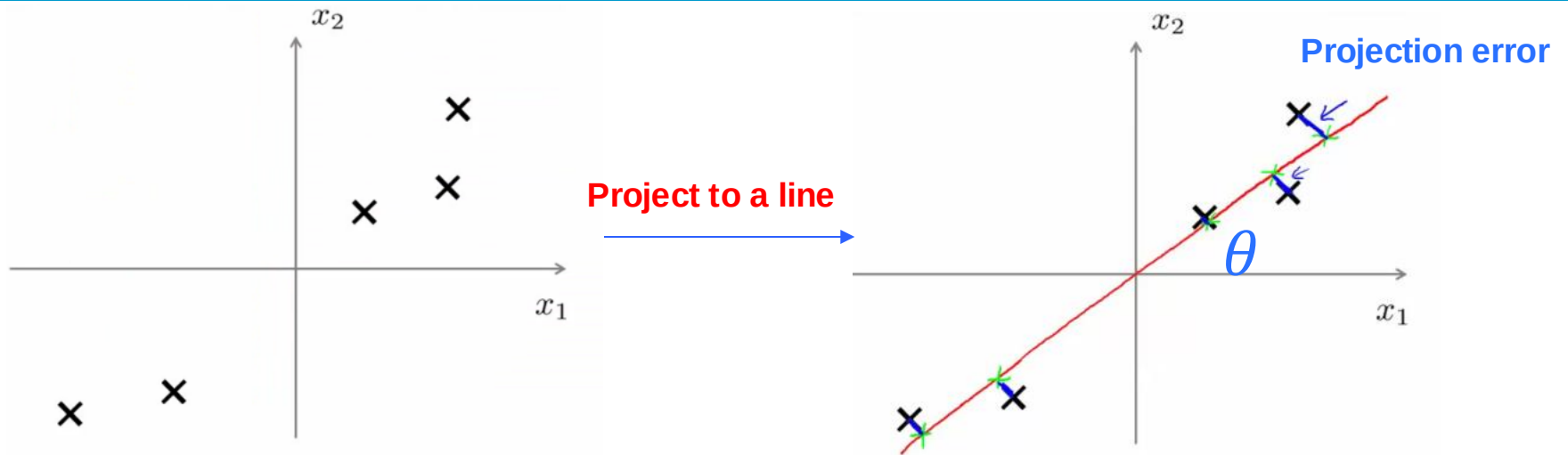
PCA



Country	X(1) GDP (trillions of US\$)	X(2) Per capita GDP (thousands of intl. \$)	X(3) Human Development Index	X(4) Life expectancy	X(5) Poverty Index (Gini as percentage)	X(6) Mean household income (thousands of US\$)	...
Canada	1.577	39.17	0.908	80.7	32.6	67.293	...
China	5.878	7.54	0.687	73	46.9	10.22	...
India	1.632	3.41	0.547	64.7	36.8	0.735	...
Russia	1.48	19.84	0.755	65.5	39.9	0.72	...
Singapore	0.223	56.69	0.866	80	42.5	67.1	...
USA	14.527	46.86	0.91	78.3	40.8	84.3	...
...	...	...	...	...	...	...	...

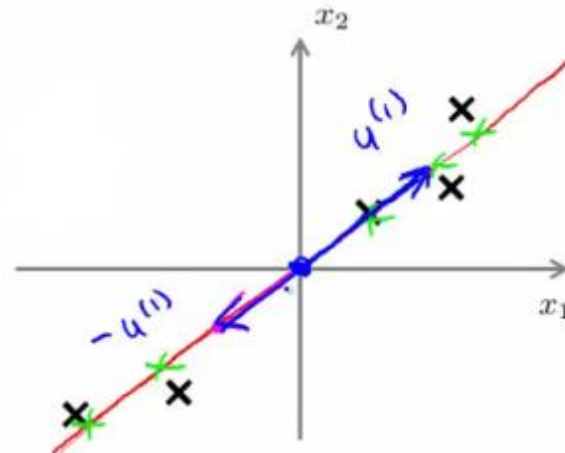
Country	z_1	z_2
Canada	1.6	1.2
China	1.7	0.3
India	1.6	0.2
Russia	1.4	0.5
Singapore	0.5	1.7
USA	2	1.5
...	...	...

PCA



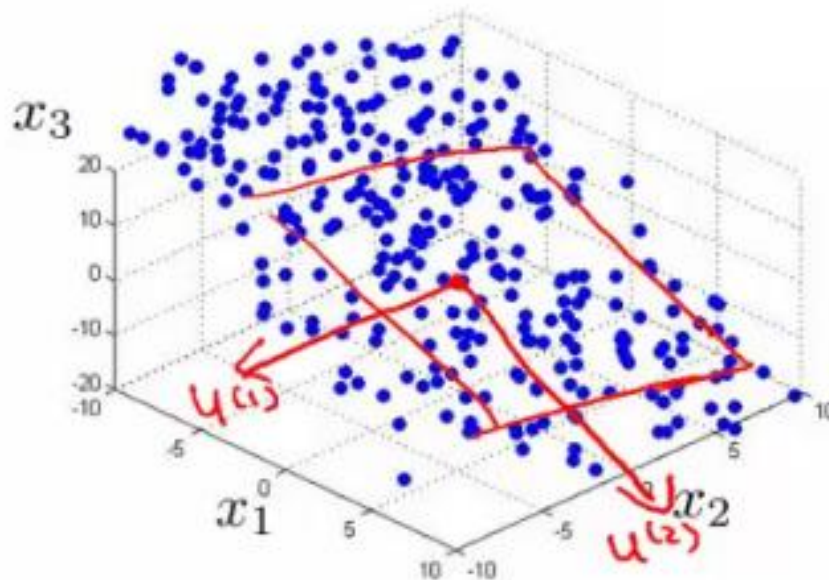
• For 2D-1D, we must find a vector $u^{(1)}$, onto which you can project the data so as to minimize the projection error.

($u^{(1)}$ - $-u^{(1)}$ are the same)



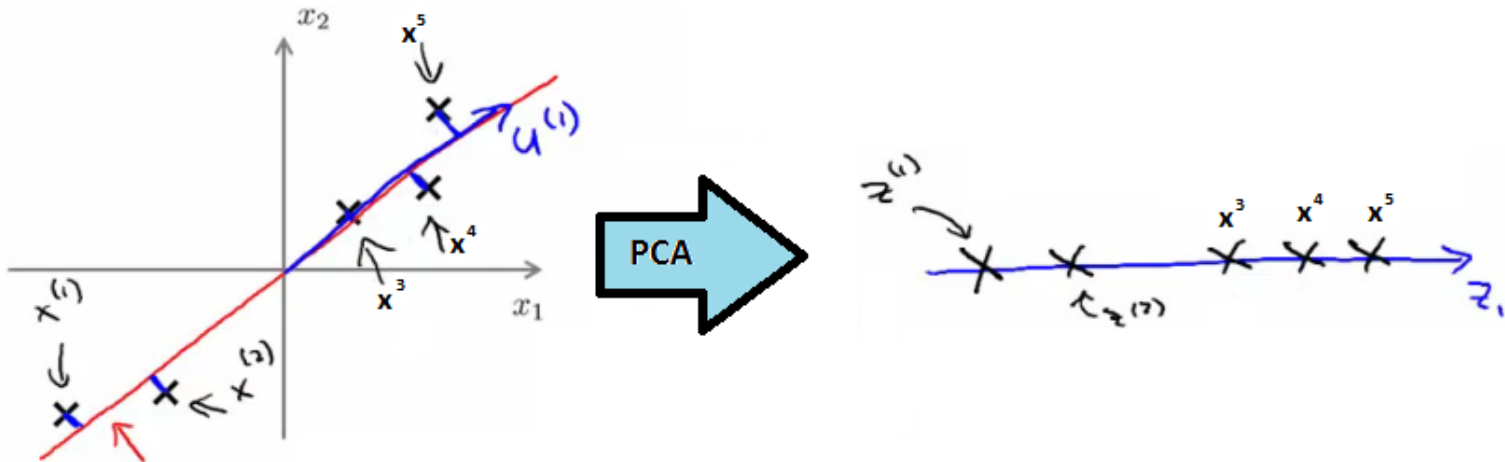
PCA

- ❖ To reduce from n dimension to k dimension, Find k vectors ($u^{(1)}, u^{(2)}, \dots, u^{(k)}$) onto which to project the data to minimize the projection error!



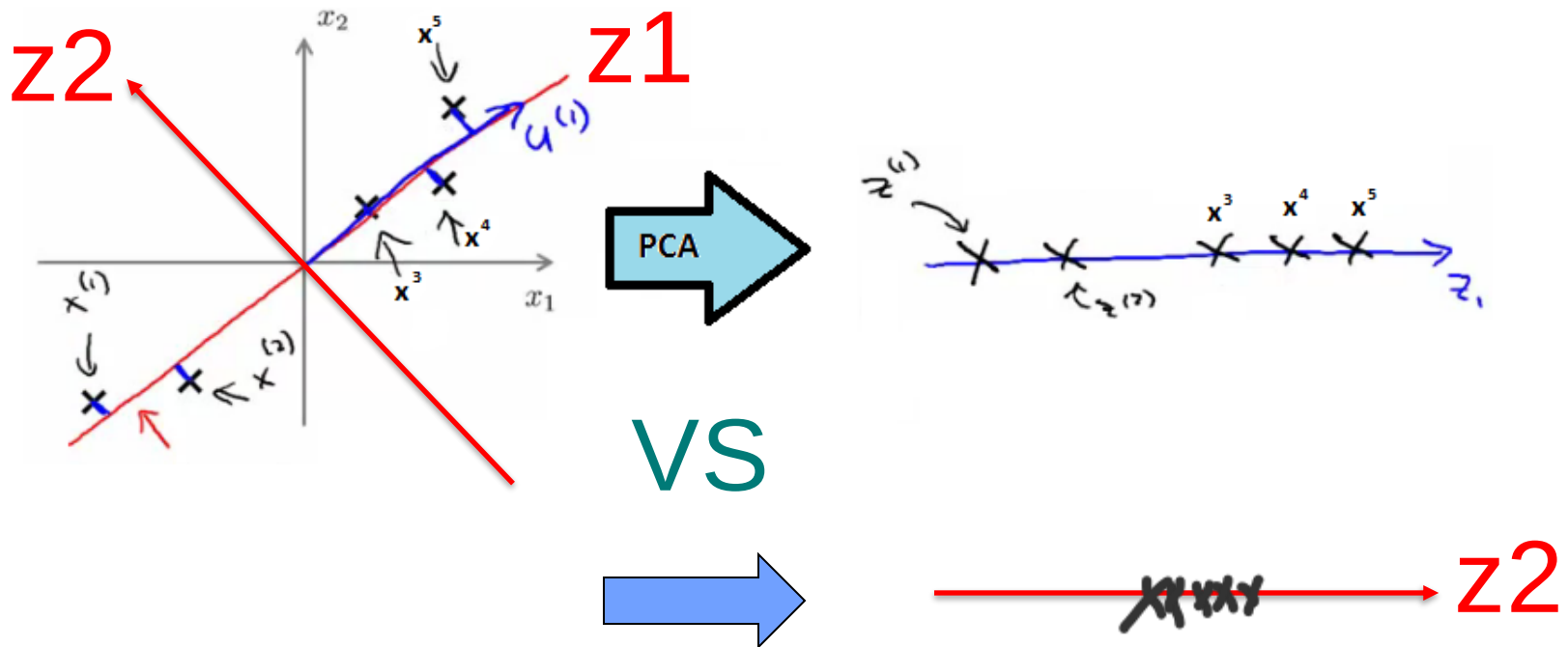
PCA

- ❖ Need to compute two things; Compute the **u vectors**
 - ⌘ The new planes
- ❖ Need to compute the **z vectors**
 - ⌘ z vectors are the new, lower dimensionality feature vectors



PCA – Importance of Principle Component

z vectors are the new, lower dimensionality feature vectors



PCA

PCA is to find a transformation of the data that satisfies the following properties:

- ❖ Each pair of new attributes has 0 covariance (for distinct attributes).
- ❖ The attributes are ordered with respect to how much of the variance of the data each attribute captures.
- ❖ The first attribute captures as much of the variance of the data as possible.
- ❖ Each successive attribute captures as much of the remaining variance as possible.

PCA-Key concepts

❖ Covariance Matrix S:

Given an m by n data matrix D , whose m rows are data objects and whose n columns are attributes, If the data matrix D is preprocessed so that the mean of each attribute is 0, then $S = D^T D$.

❖ Eigenvalues of S:

Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of S . The eigenvalues are all non-negative and can be ordered such that $\lambda_1 \geq \lambda_2 \geq \dots \lambda_{m-1} \geq \lambda_m$.

❖ Eigenvectors of S:

Let $U = [u(1), \dots, u(n)]$ be the matrix of eigenvectors of S . These eigenvectors are ordered so that the i th eigenvector corresponds to the i th largest eigenvalue.

PCA-Data Transformation

- ❖ The data matrix $D' = DU$ is the set of transformed data that satisfies the conditions posed above.
- ❖ U matrix is also an $[n \times n]$ matrix, turns out the columns of U are the u vectors we want!
 - ☞ So to reduce a system from n -dimensions to k -dimensions, just take the first k -vectors from U (first k columns)

$$U = \begin{bmatrix} | & | & & | \\ u^{(1)} & u^{(2)} & \dots & u^{(n)} \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times n}$$

U_{reduce}

$$z_1 = u^{(1)T} * x$$

$$D' = DU$$

- D is $[m \times n]$, U is $[n \times n]$
- D' is $[m \times n]$.

$$D' = D U_{\text{reduce}}$$

- D is $[m \times n]$, U_{reduce} is $[n \times k]$
- D' is $[m \times k]$.

$$z = (U_{\text{reduce}})^T * x$$

- x is $[n \times 1]$
- z is $[k \times 1]$

Use software package to Run PCA

1. Origin dataset [77 x 13]

Observations	77
Variables	13

Simple Statistics

	calories	protein	fat	sodium	fiber	carbo	sugars	potass	vitamins	shelf	weight	cups	rating
Mean	106.8831169	2.545454545	1.012987013	159.6753247	2.151948052	14.59740260	6.922077922	96.07792208	28.24675325	2.207792208	1.029610390	0.8210389610	42.66570499
StD	19.4841191	1.094789748	1.006472559	83.8322952	2.383363964	4.27895628	4.444885392	71.28681251	22.34252250	0.832524100	0.150476800	0.2327161384	14.04728874

2. Correlation matrix (it is self-standardized)

Correlation Matrix

	calories	protein	fat	sodium	fiber	carbo	sugars	potass	vitamins	shelf	weight	cups	rating
calories	1.0000	0.0191	0.4986	0.3006	-.2934	0.2507	0.5623	-.0666	0.2654	0.0972	0.6961	0.0872	-.6894
protein	0.0191	1.0000	0.2084	-.0547	0.5003	-.1309	-.3291	0.5494	0.0073	0.1339	0.2162	-.2445	0.4706
fat	0.4986	0.2084	1.0000	-.0054	0.0167	-.3180	0.2708	0.1933	-.0312	0.2637	0.2146	-.1759	-.4093
sodium	0.3006	-.0547	-.0054	1.0000	-.0707	0.3560	0.1015	-.0326	0.3615	-.0697	0.3086	0.1197	-.4013
fiber	-.2934	0.5003	0.0167	-.0707	1.0000	-.3561	-.1412	0.9034	-.0322	0.2975	0.2472	-.5131	0.5842
carbo	0.2507	-.1309	-.3180	0.3560	-.3561	1.0000	-.3317	-.3497	0.2581	-.1018	0.1351	0.3639	0.0521
sugars	0.5623	-.3291	0.2708	0.1015	-.1412	-.3317	1.0000	0.0217	0.1251	0.1004	0.4506	-.0324	-.7597
potass	-.0666	0.5494	0.1933	-.0326	0.9034	-.3497	0.0217	1.0000	0.0207	0.3607	0.4163	-.4952	0.3802
vitamins	0.2654	0.0073	-.0312	0.3615	-.0322	0.2581	0.1251	0.0207	1.0000	0.2993	0.3203	0.1284	-.2405
shelf	0.0972	0.1339	0.2637	-.0697	0.2975	-.1018	0.1004	0.3607	0.2993	1.0000	0.1908	-.3353	0.0252
weight	0.6961	0.2162	0.2146	0.3086	0.2472	0.1351	0.4506	0.4163	0.3203	0.1908	1.0000	-.1996	-.2981
cups	0.0872	-.2445	-.1759	0.1197	-.5131	0.3639	-.0324	-.4952	0.1284	-.3353	-.1996	1.0000	-.2032
rating	-.6894	0.4706	-.4093	-.4013	0.5842	0.0521	-.7597	0.3802	-.2405	0.0252	-.2981	-.2032	1.0000

Use software package to Run PCA

3. Get principle components

- Dimension is reduced from 13 to 4
- Only four of them are having Eigenvalue greater than 1.

Eigenvalues of the Correlation Matrix

	Eigenvalue	Difference	Proportion	Cumulative
1	3.60984794	0.47136754	0.2777	0.2777
2	3.13848040	1.32174475	0.2414	0.5191
3	1.81673565	0.75444087	0.1397	0.6589
4	1.06229479		0.0817	0.7406